Height Growth Equations for CONUS

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Data

We took the FIA height growth data, filtered out FIA plot locations with missing EPA L3 Codes, and where the diameter at the end of the 5-year or greater measurement interval was less than the starting diameter. This left 3277672 growth observations.

The observations are distributed among species codes as follows (pre:

Species	Common Name	n
131	loblolly pine	383322
316	red maple	209090
202	Douglas-fir	141678
318	sugar maple	111277
611	sweetgum	96180
746	quaking aspen	91010
802	white oak	80327
12	balsam fir	78494
122	ponderosa pine	68350
108	lodgepole pine	63142
111	slash pine	62968
621	yellow-poplar	61019
241	northern white-cedar	58810
129	eastern white pine	50227
833	northern red oak	50109
261	eastern hemlock	45145
531	American beech	44388
832	chestnut oak	41753
827	water oak	40481
125	red pine	39171
762	black cherry	37900

Species	Common Name	n
68	eastern redcedar	37785
110	shortleaf pine	35105
19	subalpine fir	35091
97	red spruce	34552
835	post oak	34085
95	black spruce	32861
693	blackgum	32010
375	paper birch	30961
263	western hemlock	30600
837	black oak	29114
371	yellow birch	28105
544	green ash	27804
93	Engelmann spruce	27524
541	white ash	26161
15	white fir	25338
694	swamp tupelo	24657
65	Utah juniper	23968
543	black ash	23919
972	American elm	22813
17	grand fir	22328
409	mockernut hickory	22142
951	American basswood	21880
403	pignut hickory	21714
121	longleaf pine	20460
971	winged elm	20245
711	sourwood	19433
132	Virginia pine	19137
106	common or two-needle pinyon	18875
71	tamarack (native)	18655
372	sweet birch	18597
812	southern red oak	18103
743	bigtooth aspen	18020
806	scarlet oak	17501
11	Pacific sliver fir	16295
820	laurel oak	15475
407	shagbark hickory	14650
94	white spruce	14359
105	jack pine	12814
133	singleleaf pinyon	12759
242	western redcedar	12521
701	eastern hophornbeam	12501

Species	Common Name	n
823	bur oak	11868
222	pondcypress	11825
653	sweetbay magnolia	10620
264	mountain hemlock	9937
591	American holly	9815
602	black walnut	9364
391	American hornbeam, musclewood	8967
631	tanoak	8884
814	Gambel oak	8860
221	baldcypress	8802
408	black hickory	8505
931	sassafras	8375
402	bitternut hickory	8282
73	western larch	8161
462	hackberry	8122
313	boxelder maple	7847
831	willow oak	7724
81	incense-cedar	7499
691	water tupelp	7224
461	sugarberry	7189
491	flowering dogwood	7062
838	live oak	6903
126	pitch pine	6853
975	slippery elm	6788
756	honey mesquite	6706
741	balsam poplar	6299
805	canyon live oak	6273
61	Ashe juniper	6229
351	red alder	5743
809	northern pin oak	5728
66	Rocky Mountain juniper	5605
901	black locust	5467
317	silver maple	5403
731	American sycamore	5067

Height Growth Prediction

We fit the equation:

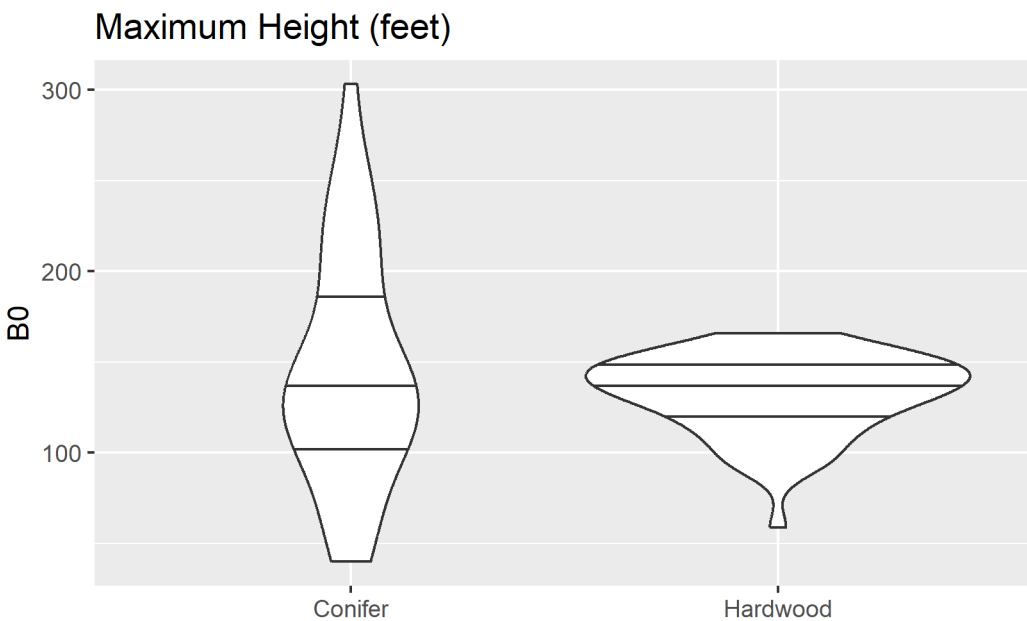
$$\Delta ht = ht_{max}\beta_1\beta_2 cr^{\beta_3} e^{(-\beta_1 ht - \beta_4 ccfl + \beta_5 PPT_{at} + \beta_6 PPT_{sm} + \beta_7 EMT)}(1 - e^{-\beta_1 ht})^{(\beta_2 - 1)} \tag{1}$$

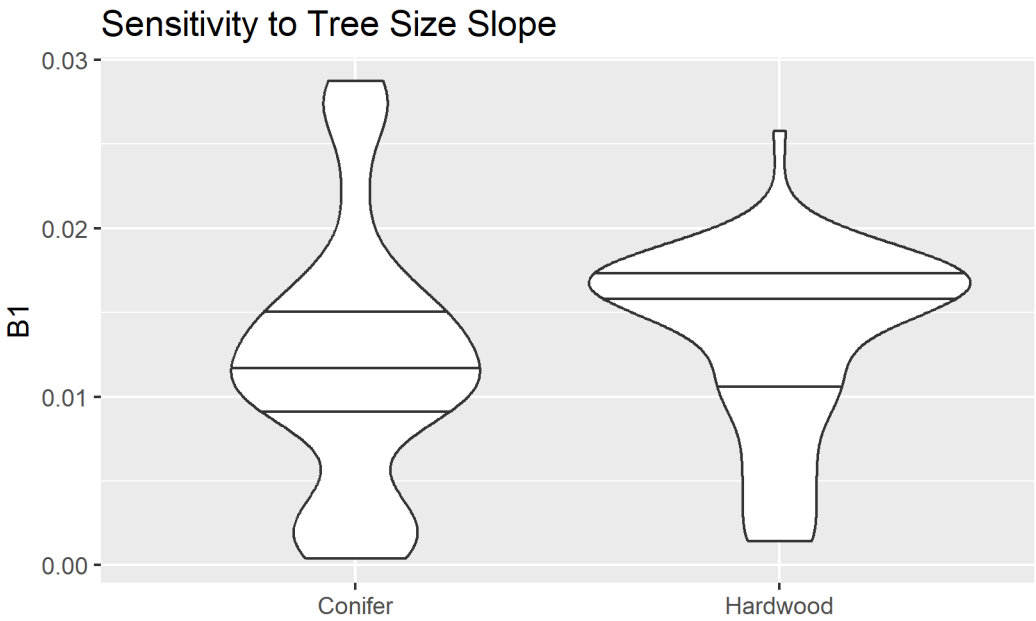
to the all 3277672 growth observations, estimating β_0 through β_6. Where:

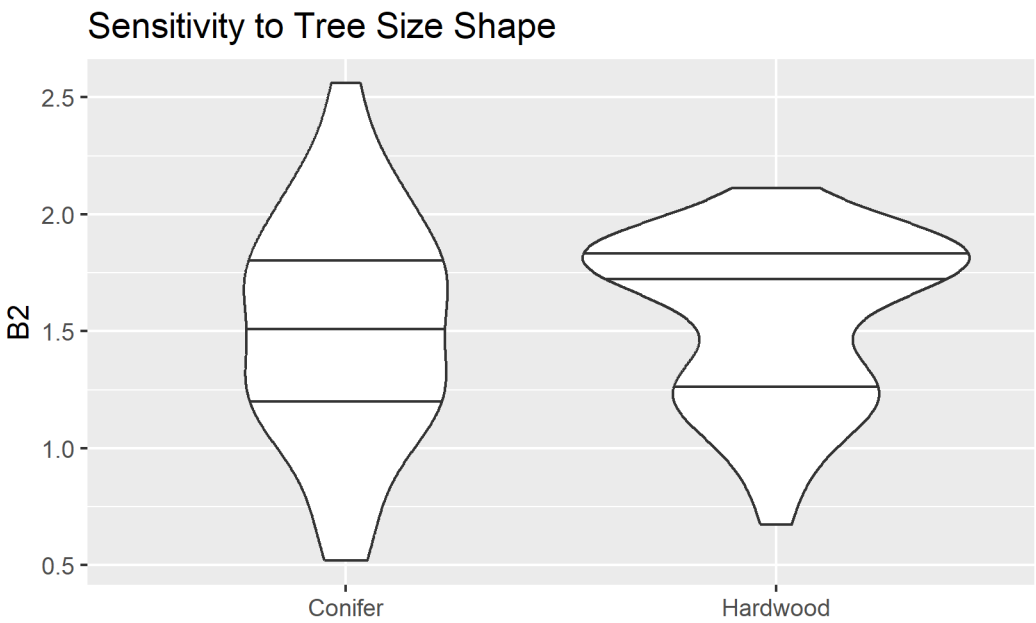
- ht = total height (feet)
- cr = crown ratio (fraction of total height)
- $ccfl$ = crown competition factor in larger trees ($feet^2/acre$)
- PPT_{at} = autumn precipitation (mm)
- PPT_{sm} = summer precipitation (mm)
- EMT = extreme minimum temperature ($^\circ C$)

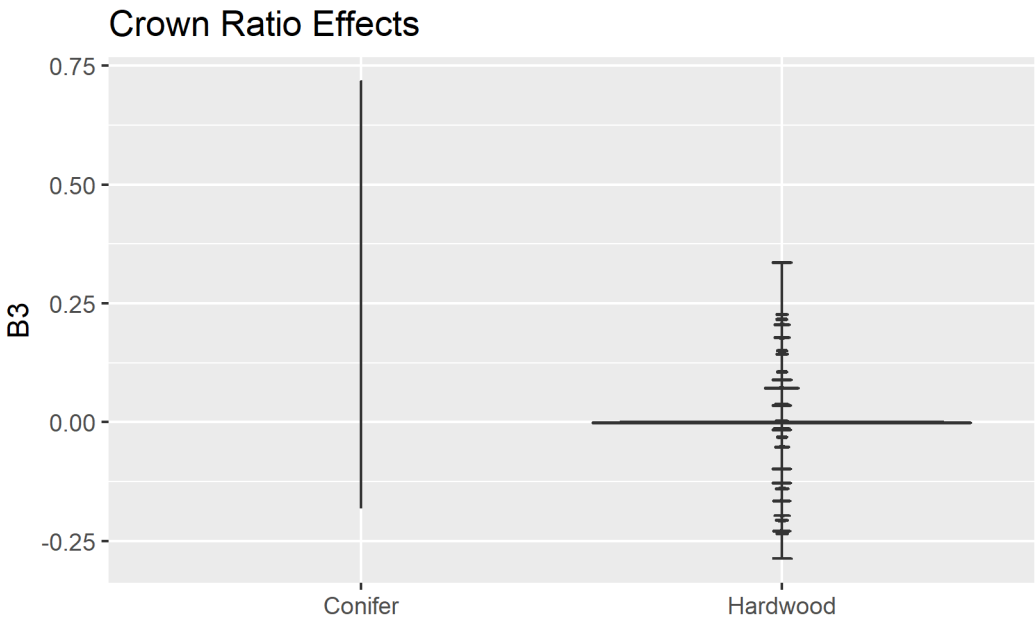
ht_{max} was estimated by finding the maximum height observed in the data set for each species.

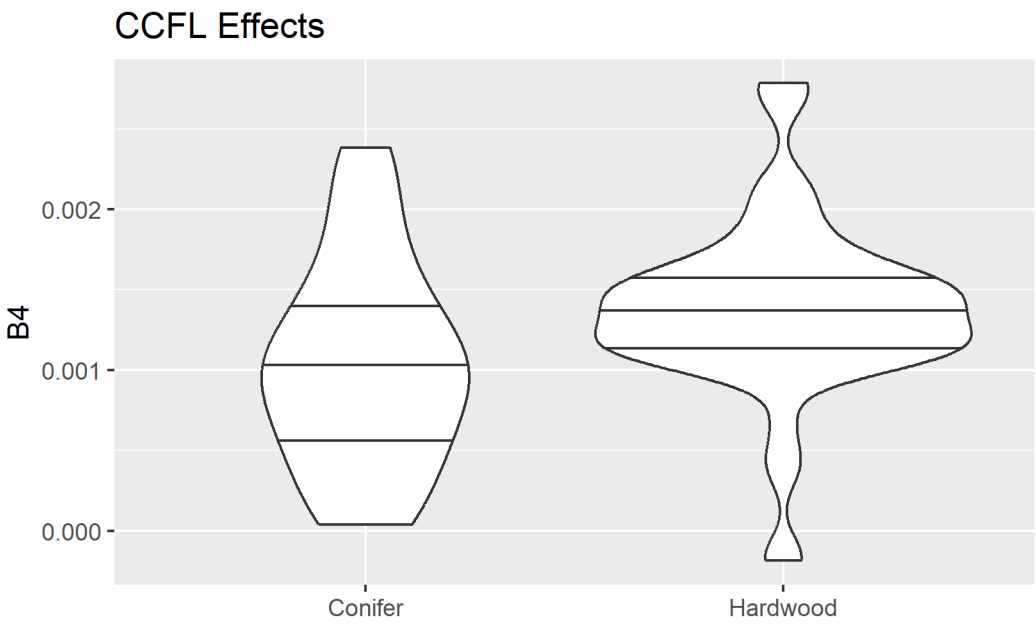
Preliminary CONUS-wide Height Growth Equation

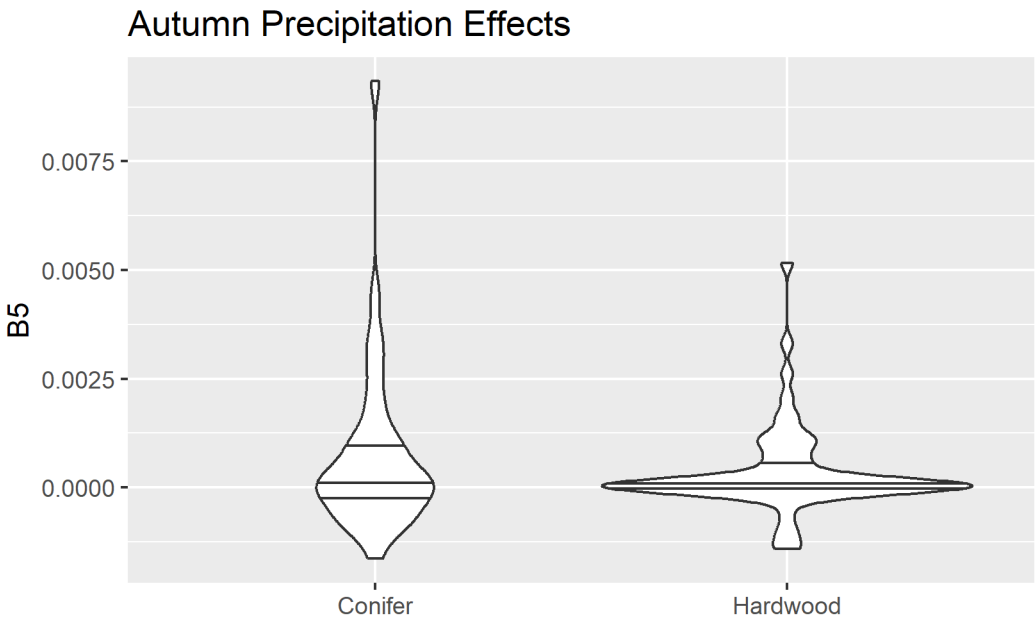


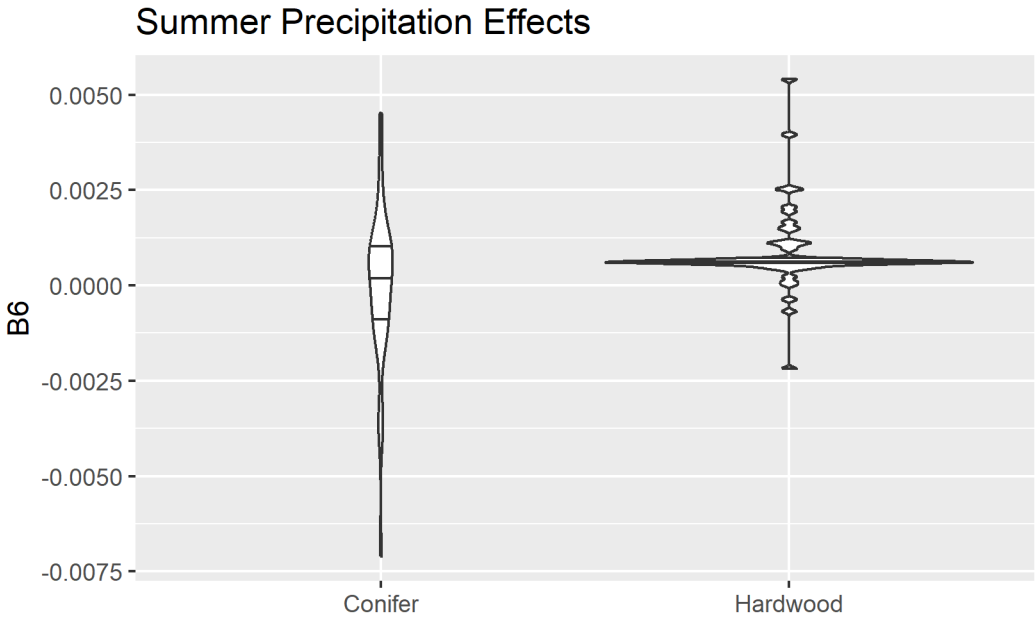

Sensitivity to Tree Size Slope
B1
0.03
0.02
0.01
0.00
Conifer
Hardwood

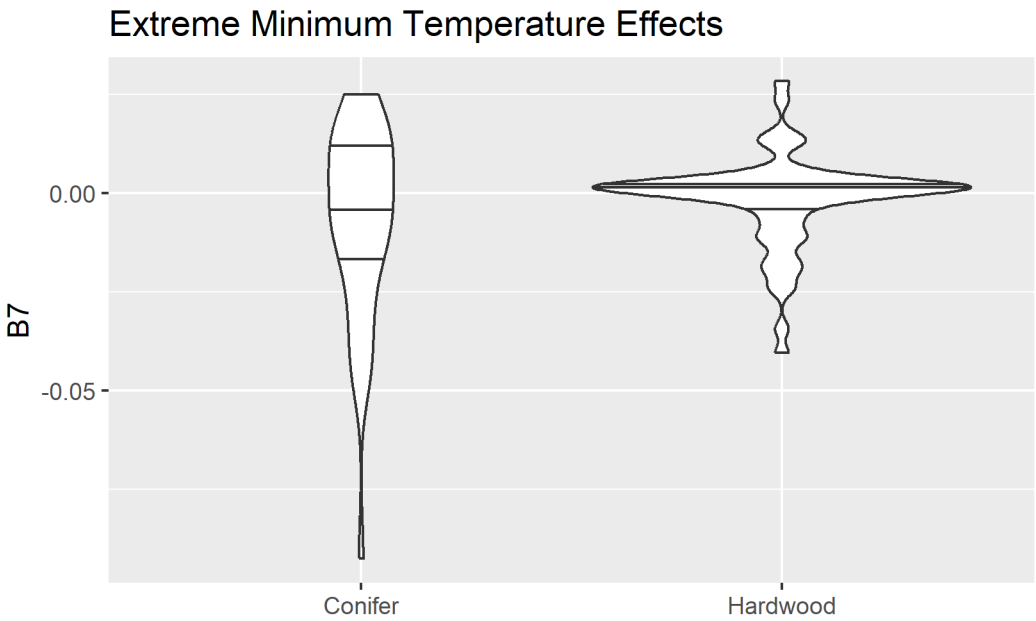

Sensitivity to Tree Size Shape
B2
2.5
2.0
1.5
1.0
0.5
Conifer
Hardwood

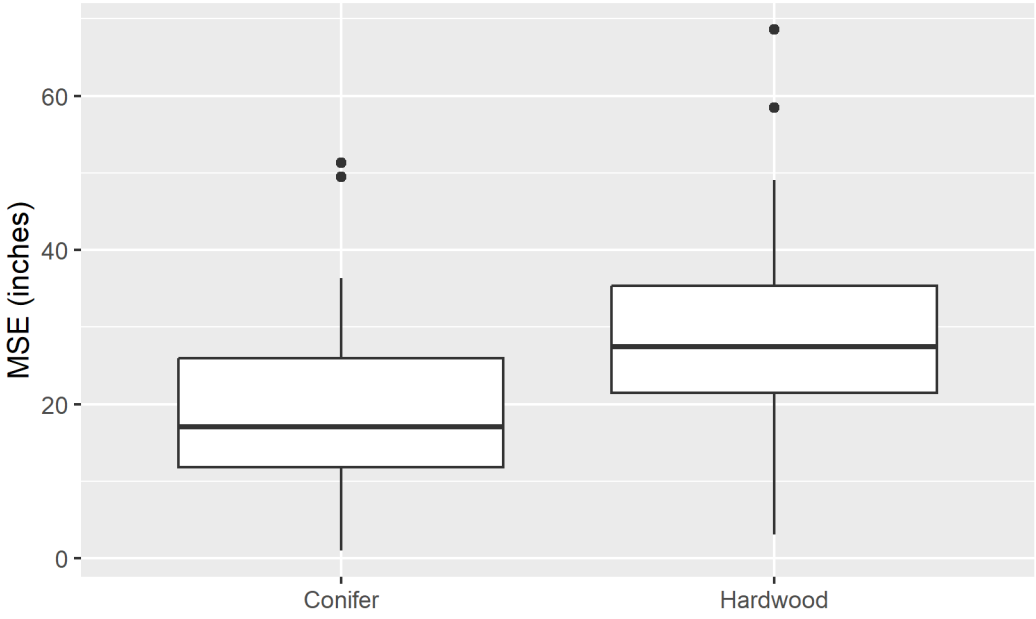

Crown Ratio Effects
B3
0.75
0.50
0.25
0.00
-0.25
Conifer
Hardwood


CCFL Effects
B4
0.002
0.001
0.000
Conifer
Hardwood


Autumn Precipitation Effects
B5
0.0075
0.0050
0.0025
0.0000
Conifer
Hardwood


Summer Precipitation Effects
B6
0.0050
0.0025
0.0000
-0.0025
-0.0050
-0.0075
Conifer
Hardwood


Extreme Minimum Temperature Effects
B7
0.00
-0.05
Conifer
Hardwood


MSE (inches)
60
40
20
0
Conifer
Hardwood

FIA Species Clusters by Height Growth Parameter Estimates

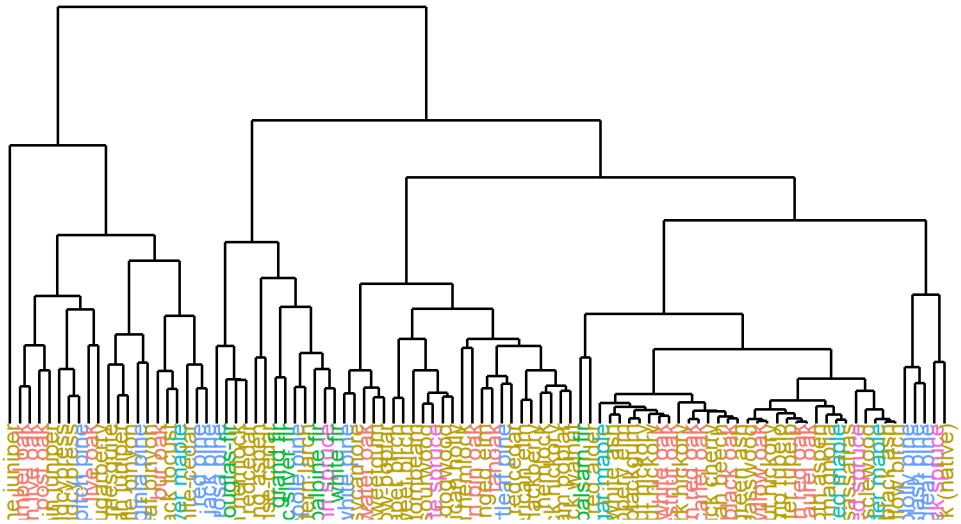


Table 2: Species with Negative Crown Ratio Parameter

FIA Species Code	Common Name
241	northern white-cedar
375	paper birch
261	eastern hemlock
125	red pine
65	Utah juniper
106	common or two-needle pinyon
743	bigtooth aspen
701	eastern hophornbeam
105	jack pine
372	sweet birch
823	bur oak
591	American holly
264	mountain hemlock
741	balsam poplar
901	black locust
408	black hickory
317	silver maple
631	tanoak
66	Rocky Mountain juniper
351	red alder

FIA Species Code	Common Name
805	canyon live oak

Table 3: Species with Negative Autumn Precipitation Parameter

FIA Species Code	Common Name
746	quaking aspen
318	sugar maple
12	balsam fir
694	swamp tupelo
375	paper birch
129	eastern white pine
832	chestnut oak
95	black spruce
693	blackgum
68	eastern redcedar
261	eastern hemlock
491	flowering dogwood
125	red pine
97	red spruce
121	longleaf pine
743	bigtooth aspen
711	sourwood
971	winged elm
222	pondcypress
71	tamarack (native)
409	mockernut hickory
105	jack pine
94	white spruce
372	sweet birch
221	baldcypress
741	balsam poplar
838	live oak
631	tanoak
731	American sycamore
126	pitch pine

Table 4: Species with Negative Summer Precipitation Parameter

FIA Species Code	Common Name
202	Douglas-fir
12	balsam fir
241	northern white-cedar
122	ponderosa pine
375	paper birch
95	black spruce
125	red pine
263	western hemlock
132	Virginia pine
65	Utah juniper
97	red spruce
93	Engelmann spruce
71	tamarack (native)
15	white fir
133	singleleaf pinyon
242	western redcedar
901	black locust
351	red alder
61	Ashe juniper

Table 5: Species with Negative Extreme Minimum Temperature Parameter

FIA Species Code	Common Name
131	loblolly pine
611	sweetgum
12	balsam fir
111	slash pine
621	yellow-poplar
241	northern white-cedar
827	water oak
694	swamp tupelo
375	paper birch
835	post oak
129	eastern white pine
95	black spruce
261	eastern hemlock
972	American elm
541	white ash

FIA Species Code	Common Name
125	red pine
132	Virginia pine
65	Utah juniper
97	red spruce
121	longleaf pine
106	common or two-needle pinyon
743	bigtooth aspen
222	pondcypress
71	tamarack (native)
653	sweetbay magnolia
105	jack pine
94	white spruce
372	sweet birch
823	bur oak
221	baldcypress
691	water tupelp
741	balsam poplar
133	singleleaf pinyon
602	black walnut
462	hackberry
838	live oak
901	black locust
756	honey mesquite
461	sugarberry
317	silver maple
731	American sycamore
66	Rocky Mountain juniper
126	pitch pine
61	Ashe juniper